

PAPER_1454_BUCKET_K_DM_DIRECT_DETECTION

Daniel T. Murphy

PAPER_1454 — UQFF: DM Direct Detection σ Floor (Bucket K)

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Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+

Tier: Bucket K

Date: June 16, 2026

Location: 41.0997° N, 80.6495° W (Youngstown, OH, USA)

Status: CLOSED — formal catalog tuple in Bucket K

Calculator surface: `calculate_bsm_constraints({...})`

Closure

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$\Lambda_{\blacksquare} \times 1e-40 \text{ cm}^2$

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NOT REPLACEMENT

Per CLAUDE.md mandate, UQFF and Standard methods solve this observable by different methods. Closure uses only canonical primitives. Zero free parameters.

Reference

- Catalog tuple resides in `_bsm_constraints_report()` or equivalent surface in `uqff_pure_calculator.py`.

- Related foundational papers: PAPER_646, PAPER_1156, PAPER_1203.

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